Multi-Level Speaker Representation for Target Speaker Extraction

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***Abstract*—Target speaker extraction (TSE) relies on a reference cue of the target to extract the target speech from a speech mixture. While a speaker embedding is commonly used as the reference cue, such embedding pre-trained with a large number of speakers may suffer from confusion of speaker identity. In this work, we propose a multi-level speaker representation approach, from raw features to neural embeddings, to serve as the speaker reference cue. We generate a spectral-level representation from the enrollment magnitude spectrogram as a raw, low-level feature, which significantly improves the model's generalization capability. Additionally, we propose a contextual embedding feature based on cross-attention mechanisms that integrate frame-level embeddings from a pre-trained speaker encoder. By incorporating speaker features across multiple levels, we significantly enhance the performance of the TSE model. Our approach achieves a 2.74 dB improvement and a 4.94% increase in extraction accuracy on Libri2mix test set over the baseline.**

***Index Terms*—Cocktail party problem, target speaker extraction, selective auditory attention, speaker feature, speaker confusion.**

I. Introduction

Target speaker extraction (TSE) aims at extracting the target speaker's speech from a mixture, which typically relies on a reference cue of the target, such as voiceprint [1]–[8], facial information [9]–[12], or even body movements [13]. Audio-only TSE relies on a pre-recorded speech to extract the target speech. Therefore, an appropriate speaker feature representation for the target speaker becomes crucial for accurate target speech extraction. Failing to extract appropriate speaker features may lead to unexpected TSE outputs, such as broken utterances or incorrect speaker output, that is called speaker confusion problem [2], [14].

The popular TSE models are jointly trained with an auxiliary module which generates the speaker embedding of the target speaker [15]–[17]. However, such jointly trained speaker encoder lacks sufficient discrimination ability between speakers [18] as far as TSE is concerned. Zhao et al. [14] attribute the speaker confusion problem to the high acoustic similarity and limitation of speaker encoder for speaker characterization. To overcome the speaker confusion problem, Mu et al. [19] propose a self-supervised disentangled representation learning to remove the local semantic information considered irrelevant to speaker identity.

Different from the joint training method, early TSE models employ pre-trained speaker models trained for the speaker verification task to provide speaker embedding [1]. Although the speaker encoder trained on numerous speakers can produce embedding with high speaker discrimination, such speaker embedding does not perform satisfactorily on the TSE task due to bad generalization [20]. Liu et al. [2], [21] attempt to solely train speaker encoders on TSE data, achieving better performance on in-domain datasets, while the encoder still exhibits limited speaker discrimination, raising concerns about the cross-domain effectiveness.

Recent studies started to explore the use of frame-level embedding generated by speaker encoders, revealing that not only speaker identity information but also contextual information from the enrolled speech, despite different linguistic content, is beneficial for the TSE task [22], [23]. Moreover, some even skipped the use of speech encoders entirely, opting to fuse speaker information directly at the spectrogram level [24], [25]. The success of these methods suggests that raw, low-level speech information, without explicit extraction of speaker identity, seems to be sufficient for effective speaker extraction. Additionally, there has been a growing interest in leveraging large self-supervised speech models to construct TSE models [26], [27]. The speaker features derived from these large self-supervised learning models have also contributed to improved TSE performance.

To investigate the practical contributions of speaker features at different granularity for TSE, we categorize speaker features into three levels based on the extent of speaker information extraction, from raw features to neural features. These levels include low-level spectral features, frame-level embeddings extracted by the speaker encoder, and utterance-level speaker embeddings derived to represent speaker identity. For low-level spectral features, we introduce a method to generate a representation from the enrollment magnitude spectrogram, referred to as the TF Map feature. The Contextual Embedding feature, which integrates raw contextual information with frame-level embeddings, is generated using a cross-attention mechanism that produces time-varying speaker embeddings

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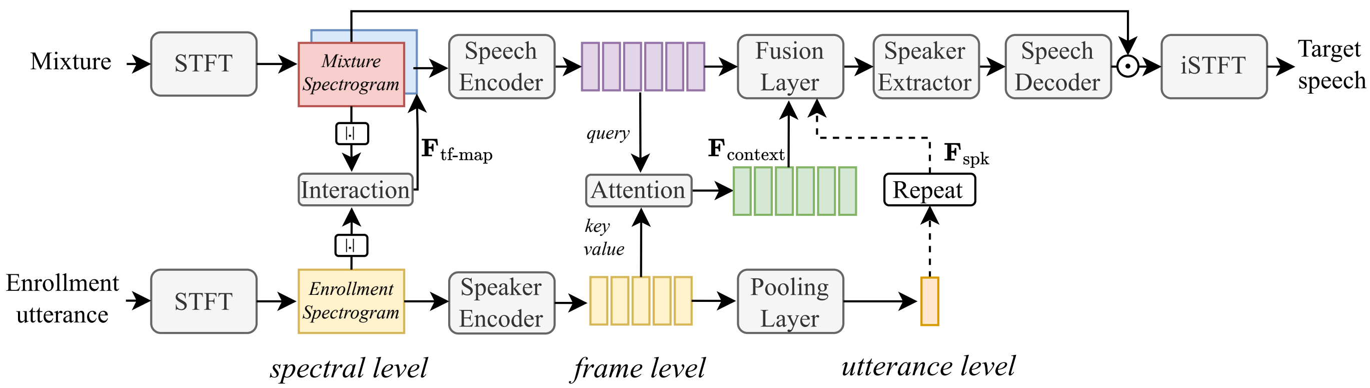

Fig. 1. The architecture of the proposed target speaker extraction model with multi-level speaker features as the reference cue. The model consists of two main pipelines. The upper left-to-right pipeline is the speaker extraction module, while the lower left-to-right pipeline is a speaker encoder that extracts multi-level speaker representation. The lower pipeline serves as the reference cue of the upper pipeline in a target speaker extraction task. $|\cdot|$ denotes the magnitude operation. $\mathbf{F}_{\text{tf-map}}$, $\mathbf{F}_{\text{context}}$, and $\mathbf{F}_{\text{spk}}$ denote the TF Map feature, Contextual Embedding feature, and Speaker Embedding feature, respectively.

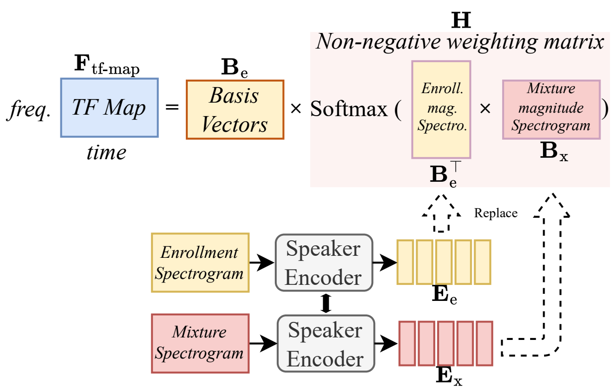

Fig. 2. The calculation process of the TF Map feature. It consists of two non-negative components: basis vectors from the enrollment's magnitude spectrogram and a weighting matrix, computed based on either 1) Spectral Similarity or 2) Embedding Similarity between the mixture and the enrollment.

for fusion. Finally, utterance-level neural speaker embedding serves as the high-level feature.

This paper is organized as follows. In Section II, we discuss the details of multi-level speaker features. In Section III, we report the experiment setup. In Section IV, we summarize the results. Finally, Section V concludes the discussion.

II. Proposed Method

We discuss three-level speaker features in conjunction with a typical speaker extraction pipeline. The Band-Split RNN (BSRNN) [28] model is selected as the backbone for the speech extractor, and the pre-trained Ecapa-TDNN [29] is employed as the speaker encoder for extracting the speaker features. Fig. 1 is the diagram of an overall architecture.

A. TF Map Feature

Since the TSE model identifies the target speaker by learning the relationship between auxiliary information and the target speech component, establishing a strong correlation between the two is essential. Spectral components from the same speaker tend to exhibit similarities, such as shared spectral structures and harmonic content [30], [31], forming a natural connection between the enrollment and target speech within the mixture. This may provide an advantage over relying solely on high-level features like speaker embedding. Therefore, we investigate whether well-constructed spectral features can serve as effective cues for guiding the TSE model to extract the target speech.

Inspired by the non-negative matrix factorization (NMF) [32]–[34] for speech reconstruction, we propose a method to generate a spectral-level representation based on the enrollment magnitude spectrogram, referred to as the TF Map feature. The generated speaker feature at the spectral level is concatenated with the spectrogram of the mixture during the extraction process, enhancing the target speech extraction.

The NMF method decomposes the magnitude spectrogram of clean speech to obtain non-negative basis vectors for individual speakers. These basis vectors are then iteratively combined through non-negative weighted summation to reconstruct the target speech in the mixture. Typically, the number of basis vectors is kept smaller than the feature dimensions, allowing the basis vectors to capture the intrinsic underlying spectral characteristics while minimizing correlation with interfering speech.

In this work, the basis vectors are simplified to include all frames of the magnitude spectrogram from the pre-enrolled utterance, and the non-negative weighting matrix is directly calculated based on the similarity using an attention mechanism, which is generated through linear correlation and a *softmax* operation. Fig. 2 illustrates the workflow of the TF Map feature.

$$\mathbf{F}_{\text{tf-map}} = \mathbf{B}_{\text{e}}\mathbf{H} \tag{1}$$

where $\mathbf{B}_{\text{e}} \in \mathbb{R}^{D_{\text{f}} \times T_{\text{e}}}$ denotes the magnitude spectrogram of the enrollment, and $\mathbf{H} \in \mathbb{R}^{T_{\text{e}} \times T_{\text{x}}}$ denotes the non-negative weighting matrix. D_{f} denotes the dimension of frequency, T_{e} and T_{x} denote the frame length of enrollment and mixture.

Here we provide two approaches for calculating the non-negative weighting matrix based on the similarity between the mixture and enrollment:

1) Spectral Similarity: The similarity between spectral frames is used to compute the non-negative weighting matrix, and then normalized using a Softmax function:

$$\mathbf{H} = \text{Softmax}(\mathbf{B}_{\mathbf{e}}^{\top}\mathbf{B}_{\mathbf{x}}) \tag{2}$$

where $\mathbf{B}_{\text{x}} \in \mathbb{R}^{D_{\text{f}} \times T_{\text{x}}}$ denotes the magnitude spectrogram of the mixture. $^{\top}$ denotes the transpose operation.

2) Embedding Similarity: To more accurately assess the similarity between the mixture's components and the basis vectors from the enrollment, both the mixture and enrollment are transformed into the embedding space through the speaker encoder:

$$\mathbf{H} = \text{Softmax}(\mathbf{E}_{\mathbf{e}}^{\top}\mathbf{E}_{\mathbf{x}}) \tag{3}$$

where $\mathbf{E}_{\text{e}} \in \mathbb{R}^{D_{\text{e}} \times T_{\text{e}}}$ and $\mathbf{E}_{\text{x}} \in \mathbb{R}^{D_{\text{e}} \times T_{\text{x}}}$ denote the sequence of the frame-level embeddings from the enrollment and the mixture, respectively. D_{e} denotes the feature dimension of the frame-level embedding.

All vectors are length-normalized when employed in calculating the weighting matrix, and the cosine similarity is used for the comparison. Besides, the energy of the constructed TF Map feature is recovered by projecting the amplitude spectrogram of the mixed signal onto the corresponding frame of the feature.

B. Contextual Embedding Feature

The speaker encoder further projects the enrollment spectrogram into an embedding space to characterize the speaker.

Frame-level embeddings can be viewed as speaker embeddings with certain contextual information, retaining the length and variability along the time axis. This characteristic provides significant flexibility when applied to TSE tasks. In this work, we employ a cross-attention mechanism to generate a time-varying speaker feature at the embedding level, referred to as Contextual Embedding. This feature maintains the same length as the spectral frames of the mixture, enabling the adaptive integration of speaker information:

$$\mathbf{F}_{\text{context}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V} \tag{4}$$

$$\mathbf{Q} = \mathbf{X}^{\top}\mathbf{W}^{\text{Q}} \tag{5}$$

$$\mathbf{K} = \mathbf{E}_{\text{e}}^{\top}\mathbf{W}^{\text{K}} \tag{6}$$

$$\mathbf{V} = \mathbf{E}_{\text{e}}^{\top}\mathbf{W}^{\text{V}} \tag{7}$$

where $\mathbf{X} \in \mathbb{R}^{D_{\text{x}} \times T_{\text{x}}}$ is the encoded mixture speech. D_{x} is the feature dimension. $\mathbf{W}^{\text{Q}} \in \mathbb{R}^{D_{\text{x}} \times d_k}$, $\mathbf{W}^{\text{K}} \in \mathbb{R}^{D_{\text{e}} \times d_k}$, and $\mathbf{W}^{\text{V}} \in \mathbb{R}^{D_{\text{e}} \times d_k}$ are learnable parameters for Query, Key and Value in attention module, respectively. d_k is the dimension in the attention module.

C. Speaker Embedding Feature

Utterance-level speaker embedding, the most common and widely used form of speaker information, is typically extracted by a speaker encoder and a pooling layer, which is primarily focused on capturing the target speaker's identity information. Due to the lack of temporal variability, the speaker embedding is replicated along the time axis to align with the speech mixture representation, resulting in the Speaker Embedding feature $\mathbf{F}_{\text{spk}}$.

In this work, we employ a multiplicative approach to construct the speaker information fusion layer, facilitating the integration of speaker identity information across the temporal dimension.

III. Experimental Setup

A. Dataset

We conduct experiments using clean Libri2mix [35] dataset, consisting of mixtures of two speakers. The TSE models are trained with 251 speakers on Libri2mix-100 containing 13900 utterances. The validation and test set each contains 3000 utterances and 40 non-overlapping speakers. Every sample is used twice to extract two different speakers by switching the enrollment. Fully overlap (min. version) and 16 kHz sampling rate are configured.

B. Training Details

The BSRNN [28] is employed as the speech extractor in the TSE model. In the speech encoder, we split the frequency band below 1.5 kHz by a 100 Hz bandwidth, the frequency band between 1.5 kHz and 3.5 kHz by a 200 Hz bandwidth, the frequency band between 3.5 kHz and 6 kHz by a 500 Hz bandwidth, treat the frequency band between 6 kHz and 8 kHz as one subband. This split results in 32 subbands. The feature dimension of each subband is 128. The speaker extractor is constructed with a 192-dimensional bidirectional LSTM repeated 6 times in the BSRNN.

The speaker encoder utilizes the Ecapa-TDNN architecture and is pre-trained with 5994 speakers on VoxCeleb2 [36]. The model is available in the open-source project WeSpeaker [37][1].

We train all TSE models for 150 epochs on segments with 3 seconds. Adam optimizer is used. The learning rate is set to $1e^{-3}$ and decays to $2.5e^{-5}$. Negative scale-invariant signal-to-noise ratio is selected as the loss function.

All the codes are implemented in PyTorch and will be released in the open source project WeSep [38] [2].

C. Evaluation Metrics

The extraction performance is evaluated through the well-known metric: the SI-SDR improvement (SI-SDRi) [39]. Besides, we consider the samples with SI-SDRi larger than 1 dB as the successful extraction. The percentage of these samples is measured for the accuracy of extraction.

[1]https://github.com/wenet-e2e/wespeaker
[2]https://github.com/wenet-e2e/wesep

TABLE I

THE RESULTS OF BSRNN WITH DIFFERENT SPEAKER FEATURES.

Model	TF Map	Contextual Embedding	Speaker Embedding	SI-SDRi / dB	Accuracy / %
BSRNN	Spec.			15.18	96.60
	Emb.			15.25	96.83
		✓		14.51	95.27
(baseline)			✓	13.17	92.08
	Spec.	✓		15.52	96.40
	Spec.		✓	15.40	96.07
	Emb.	✓		**15.91**	**97.02**
	Emb.		✓	15.60	96.63
	Emb.	✓	✓	15.85	96.95

"Spec." and "Emb." denote Spectral and Embedding Similarity, respectively. Accuracy is the percentage of samples with SI-SDRi > 1 dB.

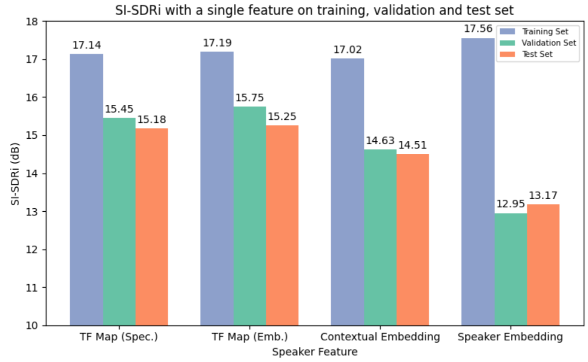

Fig. 3. SI-SDRi with a single feature on training, validation, and test set.

IV. Results and Discussion

A. Comparative Studies with Different Speaker Features

Table I shows the performance of the BSRNN with different speaker features. The model with only the speaker embedding as a speaker feature is selected as the baseline.

Among the three speaker features, the TF-map feature achieves the best performance, followed by the Contextual Embedding. The Speaker Embedding feature performs the worst, with a 2.08 dB decrease in SI-SDRi and a 4.75 percentage point drop in accuracy compared to the best one.

The use of multiple speaker features can significantly enhance the performance of TSE models. Models that incorporate multiple speaker features outperform those using a single feature. The best configuration combines both TF Map (Emb.) and Contextual Embedding features, leading to an improvement of approximately 0.7 dB in SI-SDRi and a 0.4% increase in accuracy compared to the best single-feature configuration. These findings highlight that leveraging information from different levels within the enrollment utterance is crucial for optimizing the performance of TSE models.

An interesting finding is that incorporating all three features yields performance comparable to that of using only the TF Map and Contextual Embedding features. We hypothesize that this is because the Contextual Embedding already captures the information contained in the Speaker Embedding feature.

B. Generalization of Different Features

The observation that higher-level speaker features can worsen extraction performance may seem counterintuitive. Figure 3 illustrates performance variations across the training, validation, and test sets, which may help explain this phenomenon. Higher-level features, such as the Speaker Embedding, show a larger performance gap between the training and test sets, indicating poorer generalization. These features primarily capture speaker identity, which provides a limited range of speech information and restricts data diversity available to the speaker extraction module during training. In contrast, lower-level features offer more comprehensive information beyond speaker identity, allowing the extraction module to leverage a broader data spectrum, thus enhancing performance.

TABLE II

COMPARISON WITH OTHER METHODS ON LIBRI2MIX.

Model	Speaker Model	Training Method	SI-SDRi / dB	Acc. / %	Pub.
SpeakerBeam [16]	ResNet	Joint	13.03	95.2	ICASSP, 2020
SpEx+† [15]	ResNet	Joint	13.41	-	Interspeech, 2020
sDPCCN [40]	ConvNet	Joint	11.61	-	ICASSP, 2022
Target-Conf.† [14]	ResNet	Joint	13.88	-	Interspeech, 2022
MC-SpEx† [4]	ResNet	Joint	14.61	-	Interspeech, 2023
X-T-TasNet [41]	d-vector	Pretrained	13.48	95.3	Interspeech, 2024
SSL-TD-SpeakerBeam [26]	ResNet + WavLM	Pretrained	14.01	96.1	ICASSP, 2024
		Finetuning	14.65	97.0	
BSRNN [27]	Campplus + SHuBERT	Pretrained	15.39	-	SPL, 2024
BSRNN	Ecapa-TDNN	Pretrained	**15.91**	**97.0**	Proposed
BSRNN*			**17.99**	**98.6**	

†: trained under an 8kHz sampling rate.
*: trained on the Libri2mix-360.

C. Comparison with Other Methods

Table II compares our results with other TSE models on the Libri2mix dataset. Recent studies [26], [27] explored the extraction of speaker features using large models trained through self-supervised learning (SSL). These approaches leverage not only pre-trained speaker encoders but also SSL-based speech models to extract speaker information. With the vast data and powerful encoding capabilities of SSL models, TSE models can utilize embeddings at various levels, tailored to the specific needs of the task, resulting in enhanced performance.

In contrast, our proposed method relies solely on the pre-trained ECAPA-TDNN model, providing a distinct advantage in terms of overall model size. Despite this, the strong performance of our approach shows that effectively utilizing information at different levels of the enrollment utterance alone can achieve competitive results.

V. Conclusion

We validate the effectiveness of three levels of speaker features in target speaker extraction. We confirm that integrating low-level features with speaker identity information significantly improves performance. This approach enhances generalization and mitigates overfitting, resulting in more accurate extraction outcomes.

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